Math 54 — Week 4

Matrix operations; inverses; the invertible matrix theorem; determinants (Lay 2.1–2.3, 3.1)

§1 Facts used from Weeks 1–3

(W1) Row operations are reversible and do not change solution sets.

(W2) For an $m \times n$ matrix A: pivot in every row $\iff$ columns span $\mathbb{R}^m$ $\iff$ $A\mathbf{x} = \mathbf{b}$ solvable for every $\mathbf{b}$ $\iff$ $\mathbf{x} \mapsto A\mathbf{x}$ onto. Pivot in every column $\iff$ columns independent $\iff$ $A\mathbf{x} = \mathbf{0}$ only trivially $\iff$ $\mathbf{x} \mapsto A\mathbf{x}$ one-to-one.

(W3) A linear T has standard matrix $[\,T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)\,]$. $A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v}$, $A(c\mathbf{u}) = cA\mathbf{u}$.

§2 This week

§2.1 Matrix operations (Lay 2.1)

- a_{ij} is the entry in row i, column j. The **diagonal entries** are $a_{11}, a_{22}, \ldots$. The $m \times n$ **zero matrix** is 0.
- Sum $A + B$ (same size) and scalar multiple rA are entry by entry. They satisfy the usual rules: $A + B = B + A$, $(A + B) + C = A + (B + C)$, $A + 0 = A$, $r(A + B) = rA + rB$, $(r + s)A = rA + sA$, $r(sA) = (rs)A$.

The question. If B sends $\mathbf{x}$ to $B\mathbf{x}$ and A then sends that to $A(B\mathbf{x})$, is the combined move again given by a single matrix? Which one?

Definition. If A is $m \times n$ and $B = [\,\mathbf{b}_1 \cdots \mathbf{b}_p\,]$ is $n \times p$, then $AB = [\,A\mathbf{b}_1 \;\; A\mathbf{b}_2 \;\; \cdots \;\; A\mathbf{b}_p\,]$, an $m \times p$ matrix.

The idea

The definition is chosen so that $(AB)\mathbf{x} = A(B\mathbf{x})$. Write $B\mathbf{x} = x_1\mathbf{b}_1 + \cdots + x_p\mathbf{b}_p$. By linearity, $A(B\mathbf{x}) = x_1A\mathbf{b}_1 + \cdots + x_pA\mathbf{b}_p$, which is the matrix with columns $A\mathbf{b}_j$ times $\mathbf{x}$. So multiplying matrices is composing the transformations: first B, then A. Everything else about matrix products follows from this.

- **Sizes**: $(m \times n)(n \times p) = m \times p$. The inner sizes must agree.
- **Row-column rule**: $(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$ (row i of A dotted with column j of B). Example: $\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 & 0 \\ -2 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 7 & -5 \\ -5 & 7 & 15 \end{bmatrix}$.
- Row i of AB is (row i of A) B.
- **Properties**: $A(BC) = (AB)C$; $A(B + C) = AB + AC$; $(B + C)A = BA + CA$; $r(AB) = (rA)B = A(rB)$; $I_mA = A = AI_n$. Associativity holds because both sides are the composition "first C, then B, then A".
- A^k means $A \cdots A$ (k factors); $A^0 = I$.

Three warnings.

(1) $AB \neq BA$ in general. With $C = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$:

$$CD = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}, \qquad DC = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}.$$

(2) $AB = AC$ does not imply $B = C$.

(3) $AB = 0$ does not imply $A = 0$ or $B = 0$: $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Transpose. A^T has the rows of A as its columns: $(A^T)_{ij} = a_{ji}$. Rules: $(A^T)^T = A$, $(A+B)^T = A^T + B^T$, $(rA)^T = rA^T$, and

$$(AB)^T = B^T A^T.$$

Why the order reverses: entry (i,j) of $(AB)^T$ is entry (j,i) of AB, which is (row j of A)·(column i of B) = (row i of B^T)·(column j of A^T), entry (i,j) of $B^T A^T$.

§2.2 The inverse of a matrix (Lay 2.2)

The question. For numbers, $ax = b$ is solved by $x = a^{-1}b$ when $a \neq 0$. When can $A\mathbf{x} = \mathbf{b}$ be solved the same way, and how do we compute A^{-1}?

Definition. An $n \times n$ matrix A is **invertible** if there is an $n \times n$ matrix C with $CA = I$ and $AC = I$. Then C is unique (if B also works, $B = BI = B(AC) = (BA)C = IC = C$) and is written A^{-1}. A matrix that is not invertible is **singular**.

Theorem (The 2×2 inverse)
$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible exactly when $ad - bc \neq 0$, and then $A^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. The number $ad - bc$ is the **determinant** $\det A$.

Proof. Multiplying out, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = (ad-bc)I$, and the same in the other order. If $ad - bc \neq 0$, divide by it. If $ad - bc = 0$, the same product says $A\mathbf{x} = \mathbf{0}$ for each column $\mathbf{x}$ of $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. If $A \neq 0$, one of those columns is nonzero, so $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution; an invertible A cannot have one, since $\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$. If $A = 0$ it is clearly not invertible. □

Theorem (Invertible matrices give unique solutions)
If A is invertible, then for each $\mathbf{b}$ the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

Proof. $A(A^{-1}\mathbf{b}) = \mathbf{b}$, so it is a solution. If $A\mathbf{u} = \mathbf{b}$, multiply by A^{-1}: $\mathbf{u} = A^{-1}A\mathbf{u} = A^{-1}\mathbf{b}$. □

Example: solve $3x_1 + 4x_2 = 3$, $5x_1 + 6x_2 = 7$.

$$\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}^{-1} = \frac{1}{18-20}\begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}\begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

Theorem (Algebra of inverses)
If A, B are invertible $n \times n$: $(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$; $(A^T)^{-1} = (A^{-1})^T$.

Proof. Check the definition. $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$, and similarly in the other order. $A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I$, using $(XY)^T = Y^TX^T$. □

The order reverses because AB means "first B, then A"; to undo it, undo A first and then B.

Elementary matrices. An **elementary matrix** E is obtained by doing one row operation to I. Then EA is the result of doing that same row operation to A (check on the rows: row i of EA is (row i of E)A). Every elementary matrix is invertible: its inverse is the elementary matrix of the reverse operation.

Theorem (Invertible $\iff$ row equivalent to I; the inverse algorithm)
An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n. In that case, any sequence of row operations that reduces A to I_n also turns I_n into A^{-1}.

The idea

Reducing A to I by row operations $E_1, \dots, E_k$ means $E_k \cdots E_1 A = I$. The product $E_k \cdots E_1$ is therefore a left inverse of A, and it is invertible (each E_i is), so it is A^{-1}. And $E_k \cdots E_1 = E_k \cdots E_1 I$ is what you get by doing the same row operations to I. Writing A and I side by side as $[\,A \;\; I\,]$ does both at once.

Proof.
1. If A is invertible, $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$, so A has a pivot in every row and every column. A square matrix with a pivot in every column reduces to I_n.
2. If $E_k \cdots E_1 A = I$, let $C = E_k \cdots E_1$. C is invertible (a product of invertible matrices), and $CA = I$ gives $A = C^{-1}CA = C^{-1}$, so A is invertible with $A^{-1} = C$.
3. $C = CI$ is the result of applying the same row operations to I.

□

Computing A^{-1}. Row reduce $[\,A \;\; I\,]$. If A reduces to I, the result is $[\,I \;\; A^{-1}\,]$. If a row of zeros appears in the left half, A is not invertible.

Worked example.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & 1 & 5 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right]$$

$$\xrightarrow[R_1 - 2R_3]{R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 2 & -2 \\ 0 & 1 & 0 & -4 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right], \qquad A^{-1} = \begin{bmatrix} -3 & 2 & -2 \\ -4 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix}.$$

Check one entry of AA^{-1}: row 2 of A times column 1 of A^{-1} is $2(-3) + 1(-4) + 5(2) = 0$, as required off the diagonal.

§2.3 The invertible matrix theorem (Lay 2.3)

Theorem (The invertible matrix theorem)

For a *square* $n \times n$ matrix A, the following are all equivalent (all true or all false):

(a) A is invertible.
(b) A is row equivalent to I_n.
(c) A has n pivot positions.
(d) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
(e) The columns of A are linearly independent.
(f) $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
(g) $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$.
(h) The columns of A span $\mathbb{R}^n$.
(i) $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
(j) There is C with $CA = I$.
(k) There is D with $AD = I$.
(l) A^T is invertible.

The idea

For a square matrix, "a pivot in every row" and "a pivot in every column" are the same statement: both say there are n pivots. That single fact merges the onto-list (g)–(i) with the one-to-one list (d)–(f) from (W2), and (b), (c) are just other names for n pivots. A one-sided inverse forces n pivots, which forces a two-sided inverse.

Proof. 1. (b) $\iff$ (c): a square matrix has n pivots exactly when its RREF is I_n. (a) $\iff$ (b) is the previous theorem.
2. (c) $\iff$ (d) $\iff$ (e) $\iff$ (f): n pivots in n columns means a pivot in every column; use (W2).
3. (c) $\iff$ (g) $\iff$ (h) $\iff$ (i): n pivots in n rows means a pivot in every row; use (W2).
4. (a)$\Rightarrow$(j) and (a)$\Rightarrow$(k): take $C = D = A^{-1}$.
5. (j)$\Rightarrow$(d): if $CA = I$ and $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = CA\mathbf{x} = C\mathbf{0} = \mathbf{0}$.
6. (k)$\Rightarrow$(g): if $AD = I$, then $\mathbf{x} = D\mathbf{b}$ solves $A\mathbf{x} = \mathbf{b}$.
7. (a) $\iff$ (l): if A is invertible, $(A^{-1})^T$ is the inverse of A^T; apply this to A^T for the converse, since $(A^T)^T = A$.

□

- **Use it in both directions.** To show A is invertible, check any one statement; to show it is not, find one that fails (two equal columns, a zero row, a free variable).
- If $AB = I$ for square A, B, then automatically $BA = I$ and $B = A^{-1}$ (by (k)$\Rightarrow$(a)).
- The theorem is only for square matrices. A 2×3 matrix can have a pivot in every row without having independent columns.
- **Invertible linear transformations.** $T\colon \mathbb{R}^n \to \mathbb{R}^n$ with standard matrix A is invertible (there is S with $S(T(\mathbf{x})) = \mathbf{x}$ and $T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x}$) exactly when A is invertible, and then $S(\mathbf{x}) = A^{-1}\mathbf{x}$.

§2.4 Introduction to determinants (Lay 3.1)

The question. For 2×2 matrices, one number $ad - bc$ decides invertibility. Is there such a number for $n \times n$ matrices?

Definition. For $n \geq 2$, let A_{ij} be the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A. The (i,j)-**cofactor** is $C_{ij} = (-1)^{i+j} \det A_{ij}$. The **determinant** is defined recursively by expanding along the first row:

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n},$$

with $\det[a] = a$ for 1×1 matrices.

For $n = 2$ this gives $a_{11}a_{22} - a_{12}a_{21} = ad - bc$. The signs $(-1)^{i+j}$ alternate like a checkerboard, starting with $+$ in the top left:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Theorem (Cofactor expansion along any row or column)

$\det A$ can be computed by expanding along any row i, $\det A = a_{i1}C_{i1} + \cdots + a_{in}C_{in}$, or any column j, $\det A = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj}$.

(Lay states this without proof.) The point is speed: expand along the row or column with the most zeros, because a zero entry a_{ij} makes the term $a_{ij}C_{ij}$ zero, so that cofactor never has to be computed.

Theorem (Triangular matrices)

If A is triangular (all entries below the diagonal are 0, or all above), $\det A$ is the product of the diagonal entries.

Proof. For upper triangular A, expand down the first column: only a_{11} is nonzero, so $\det A = a_{11} \det A_{11}$, and A_{11} is again upper triangular. Repeat. (For lower triangular, expand along the first row.) □

Example:

$$\det \begin{bmatrix} 2 & 0 & 0 \\ 7 & -3 & 0 \\ 5 & 1 & 4 \end{bmatrix} = 2 \cdot (-3) \cdot 4 = -24.$$

Next week: how row operations change det, which gives a fast method and the theorem $\det A \neq 0 \iff A$ invertible.

Shortcuts

- **Size check before multiplying**: $(m \times n)(n \times p)$. If the inner numbers differ, the product is undefined.
- **One entry of AB**: row of A dot column of B; you rarely need the whole product.
- **2×2 inverse**: swap the diagonal, negate the off-diagonal, divide by $ad - bc$.
- **Solving several systems with the same A**: row reduce $[\,A \;\; \mathbf{b}_1 \;\; \mathbf{b}_2\,]$ once.
- **Invertible or not?** For a square matrix, any one IMT statement settles it: count pivots, spot dependent columns, or compute det for 2×2.
- **Reverse order**: $(AB)^{-1} = B^{-1}A^{-1}$ and $(AB)^T = B^T A^T$.
- **Determinant by cofactors**: choose the row or column with the most zeros; mind the checkerboard signs. Triangular: multiply the diagonal.
- **Check an inverse**: multiply one row of A by a couple of columns of your A^{-1}; you should get the corresponding entries of I.

Traps

- $AB \neq BA$; you cannot cancel; $AB = 0$ with $A, B \neq 0$ happens.
- $(A + B)^2 = A^2 + AB + BA + B^2$, not $A^2 + 2AB + B^2$.
- $(AB)^{-1} \neq A^{-1}B^{-1}$ in general.
- The invertible matrix theorem applies only to square matrices.
- In $[\,A \;\; I\,]$, apply every row operation to the whole row, both halves.
- Cofactor signs: $C_{ij} = (-1)^{i+j} \det A_{ij}$; forgetting the sign is the most common determinant error.

§3 Homework this week

HW3, Section 2.1: 1, 3, 7, 9, 11, 12, 15, 17, 22, 35. Section 2.2: 1, 5, 7, 9, 12, 17, 18, 19, 20, 23, 29, 31, 33, 41. Section 2.3: 3, 7, 11, 13, 15, 21, 23, 28, 33, 38, 41, 44. HW4, Section 3.1: 1, 5, 11, 13, 21. (The rest of HW4 is in the Week 5 handout.)

§4 Problem set

P1. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Compute AB, BA, $(AB)^T$ and $B^T A^T$.

P2. Compute X^3 for $X = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$, and guess X^k.

P3. Use the 2×2 inverse to solve $3x_1 + 4x_2 = 3$, $5x_1 + 6x_2 = 7$, and then $3x_1 + 4x_2 = 1$, $5x_1 + 6x_2 = 0$.

P4. Find A^{-1} for $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix}$, and check your answer.

P5. Is $\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 0 & 1 & 5 \end{bmatrix}$ invertible? Answer with as little computation as possible.

P6. Compute $\det \begin{bmatrix} 3 & 0 & 4 & 1 \\ 2 & 0 & -1 & 0 \\ 5 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 \end{bmatrix}$ by choosing good rows and columns.

P7. True or false, with a reason (all matrices $n \times n$). (a) If $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then A has fewer than n pivots. (b) If the columns of A span $\mathbb{R}^n$, then they are linearly independent. (c) If $AB = I$, then A and B are both invertible. (d) If A is invertible, then A^2 is invertible.

P8. Suppose A is $n \times n$ and $A^3 = 0$. Show that $I - A$ is invertible, with inverse $I + A + A^2$.